

Why pre-training converges at rate $1/\sqrt{m}$

The Section 4 claim $\hat{\theta} = \theta^* + O_p(m^{-1/2})$ — and Appendix B [Nagler '23]

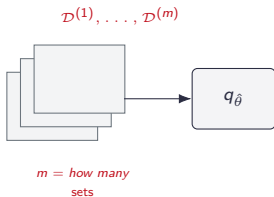
Gurmeher Singh Puri

Explainer deck — the PFN approximation result

First: m is *not* the famous size

Section 4 has **two different “sizes”** — don't conflate them in Q&A.

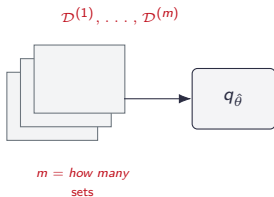
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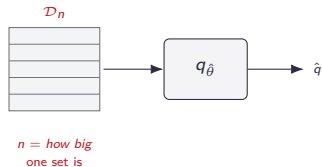
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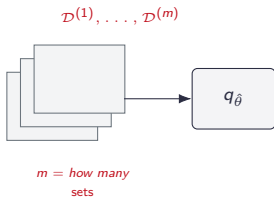
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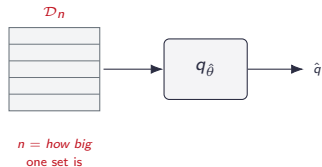
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This deck is about m — the **Monte-Carlo** count. The surprising ICL-past-1000 story is about n (Section 5).

The claim, and where it lives

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“...we can expect $\hat{\theta} = \theta^* + O_p(m^{-1/2})$,
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(not “quality grows like $\sqrt{m}$ ” — the *error* decays like $m^{-1/2}$, so $\hat{\theta} \rightarrow \theta^*$ as $m \rightarrow \infty$.)

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θ^* the KL-optimal parameter (best the model can do on the prior)

$\hat{\theta}$ what pre-training actually returns from m simulated sets

And yes — Appendix B is exactly the section this points to.

Step 1 — what is pre-training even maximizing?

An **objective** = one number that *grades* a parameter setting θ (bigger = better predictions).
Pre-training turns the knobs θ to push this grade as high as it goes. (like maximizing a test score)

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The question: does maximizing the **computable** L_m land near the maximizer of the **true** L ?
Yes — and the gap is the $O_p(m^{-1/2})$. L_m is an *average of m pieces* — the “M-estimator” setup.

Aside — what is the Central Limit Theorem (CLT)?

Take many **independent** random measurements and **average** them. The CLT says:

the average lands near the *true* mean, and its typical error **shrinks like $1/\sqrt{\text{\#measurements}}$**
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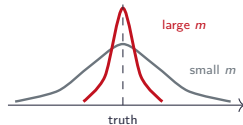
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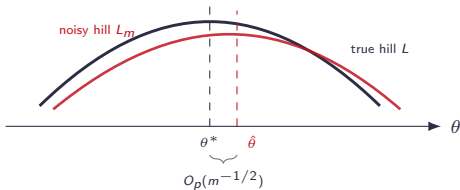
the bell narrows as $m \uparrow$



Our L_m is exactly such an average (over m datasets) — so its error, and $\hat{\theta}$'s, shrink like $1/\sqrt{m}$.

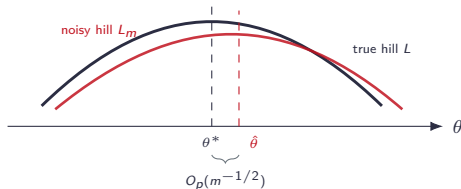
Why $\sqrt{m}$: a noisy hill has a slightly-shifted peak

Picture each objective as a **hill** over θ ; the best θ is the **peak** (slope flat). L_m is a **noisy copy** of the true hill $L \Rightarrow$ its peak $\hat{\theta}$ sits just off θ^* .



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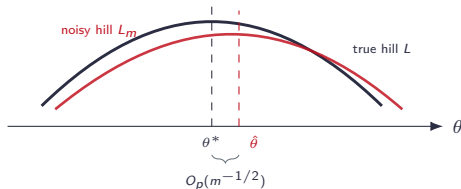
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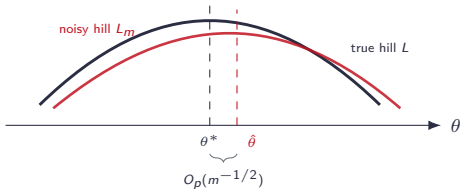
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2. That leftover slope is an **average of m independent pieces** $\Rightarrow$ by the **CLT**, $\approx 1/\sqrt{m}$ in size.
3. A small slope-error shifts the peak proportionally (less if the hill is sharply curved) $\Rightarrow$
 $\hat{\theta} - \theta^* \approx 1/\sqrt{m}$. (slope = gradient, curvature = Hessian.)

Appendix B, Thm B.1 — the parameter

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Why the sandwich (not plain inverse-Fisher)? The PFN is **misspecified** ($q_{\theta} \neq \pi$), so $V \neq -I$.

Proof = a citation: van der Vaart & Wellner (1996), Cor. 3.2.3 & Ex. 3.2.12.

Appendix B, Thm B.2 — from $\hat{\theta}$ to the prediction

Push the rate from the parameter onto the actual output $q_{\hat{\theta}}$ (continuous mapping + delta method):

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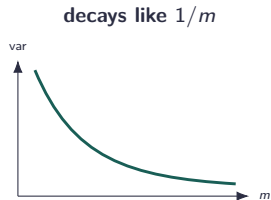
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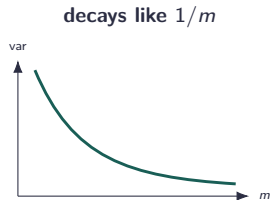
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Variance scales with **(a)** how tightly $\hat{\theta}$ is pinned (Σ_{θ^*}) and **(b)** how sensitive the output is to θ ($\nabla_{\theta} q$).

⇒ **more complex models need a larger m .**

Where this sits: the four knobs of Section 4

The PFN approximation depends on **four** factors:

data prior
 Π

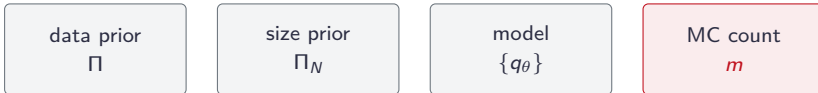
size prior
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model
 $\{q_\theta\}$

MC count
 m

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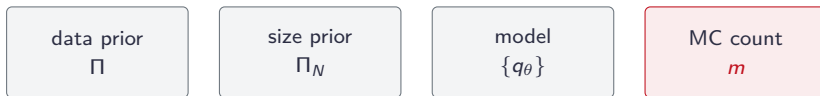
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The paper calls this axis “*rather uneventful*”: it is **orthogonal** to the in-context story (which is driven by n , not m).

Π_N (size of each set) is the “complexity regularizer”; m (**how many sets**) is this $\sqrt{m}$ rate — two separate knobs.

$\hat{\theta}$ is just an M-estimator over m iid simulated sets.

So it is $\sqrt{m}$ -consistent (Appendix B) — the *boring* axis,
orthogonal to in-context learning in n .